



BROTHER ENTERPRISES HOLDING CO., LTD.

ZHOUWANGMIAO, HAINING CITY, ZHEJIANG PROVINCE, CHINA

TEL: 86-573-87538381 FAX: 86-573-87538380

MATERIAL SAFETY DATA SHEET

Vitamin K3 MSB

1. COMPOSITION / INFORMATION ON INGREDIENT

1.1 Product Information

The product consists of Menadione Sodium Bisulfite which diluted with Sodium Bisulfite

1.2 Company identification

Manufacture Name: BROTHER ENTERPRISES HOLDING CO., LTD.,

Address: ZHOUWANGMIAO, HAINING CITY, ZHEJIANG PROVINCE, CHINA

Postcard: 314407

TEL: 0086-573-87538381

FAX: 0086-573-87538380

1.3 Phone number of Technical supporter

if you want to have more information about the product or any problem, please call this phone 0086-573-87536767

2. HAZARDS IDENTIFICATION

Very soluble in sweat, in mucosa lining of the nose, mouth and conjunctiva tissue. When in contact with skin, respiratory tract, eyes, may cause rash, itching, irritation. If not properly removed, may cause dermatitis, conjunctivitis.

Ingestion: effects or symptoms due to ingestion have not been reported. Keep in mind that the product has Vitamin K activity (it aids in the clotting of blood) and is used as anti-hemorrhagic agent.

3. FIRST AID MEASURES

Eyes: rinse with plenty of cold water while holding eyelids well open. Consult a physician if any symptoms persist.

Skin: remove contaminated clothing. Rinse skin with plenty of cold water, apply anti irritation cream.

Ingestion: rinse mouth with cold water. In order to dilute gastric contents, drink plenty of water. At the first symptoms seek medical advice and show in any case this MSDS.

Inhalation: flush nose and mouth with cold water. Remove from area of exposure. If any symptoms develop, consult a physician.

4. FIRE FIGHTING MEASURES

In case of fire, product may form irritant menadione vapours and toxic fumes due to the presence of CO and NO_x. Extinguish preferably with water spray, carbon dioxide, dry chemical, foam. All persons fighting a fire in confined spaces must wear adequate protective equipment including self-contained breathing apparatus (see sect. 10).

5. ACCIDENTAL RELEASE MEASURES

Wear individual protective equipment according to the circumstances. Collect spilled material avoiding dusts formation. do not allow product to enter drains, surface waters. Transfer in suitable containers for disposal in compliance with law.

6. HANDLING AND STORAGE

6.1 Handling

When handling apply local exhaust ventilation. Avoid dust formation. Wear protective equipment (see sect. 8) Avoid skin contact and eye contact. Wash thoroughly with cold water after handling.

6.2 Storage

Keep containers tightly closed and away from heat. Store in cool, dry and well ventilated place. Avoid heat and light exposure, store separate from basic substances. Standard packaging: carton boxes with inner polyethylene bag.

7. PERSONAL PROTECTION

7.1 Avoid dust formation.

7.2 Equipment to provide adequate personal protection

Respiratory protection: particulate dust respirator.

Hand protection: polyethylene, PVC gloves.

Eye protection: safety goggles.

8. PHYSICAL AND CHEMICAL FEATURES

Appearance: white to yellowish crystalline powder

Odour: odourless.

pH (sol 10 % in water): 4.5 --7.0

Melting point: 121-124°C

Flammability (Solids): not available

Solubility: very soluble in water (1 g in 2 ml H₂O)

9. STABILITY AND REACTIVITY

The product is stable under the suggested handling and storage conditions (see sect. 7) not causing dangerous reactions.

Materials to avoid: strong oxidising agents, strong acids, strong alkalis.

Condition to avoid: light, heat and moisture.

Hazardous decomposition products: irritant menadione vapours and fumes containing CO, CO₂, NO_x, SO_x, are produced respectively by thermal decomposition or when the product burns.

10. TOXICOLOGICAL INFORMATION

(Experimental data)

LD₅₀ (oral rat): > 2,000 mg/Kg

Skin irritation: negative determined on rabbit for similar products. Observed effects from industrial exposure suggest to consider the product irritant to skin.

Eye irritation: positive determined on rabbit for similar products.

This product is not listed as a carcinogen or potential carcinogen in IARC or OSHA. There is no evidence of mutagenic or teratogenic effects.

11. ECOLOGICAL INFORMATION

Use the product according to good manufacturing and handling practice, do not dispose improperly.

12. DISPOSAL CONSIDERATION

In any case follow indications of Sec. 7 and 8.

In case of small quantities flush with plenty of water .

In case of consistent amounts, collect the product in a close container and incinerate in qualified installation with appropriate flue gas scrubbing in compliance with the local laws and regulations.

13. TRANSPORT INFORMATION

Not Dangerous according to above mentioned rules.

14. SAFEGUARD INFORMATION

Harmful is swallowed

Irritating to eyes, respiratory system and skin.

Do not breathe dust.

Avoid contact with skin and eyes.

In case of contact with eyes, rinse immediately with plenty of water and seek medical advice.

Wear suitable gloves

15. UNDESIRABLE SUBSTANCES:

We hereby guarantee that heavy – metal content below 0.002% , which meet with European Directive 2003/10/CE. The undesirable substance is free of Arsenic, Lead, Fluorine, Mercury and Cadmium.

16. OTHER INFORMATION

16.1 APPLICATION

For use in animal feed only.

16.2 Training advice

All persons handling this product should be informed of the existence of the hazard, of any possible damage to which they might be subject and of all requisite protective measures to prevent such damage.

ORIGINAL CONTROLLED